TITLE

Socioeconomic Gradients and The Prevalence of Cardiovascular Diseases Among Older Adults in South Africa

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ABSTRACT (250 word max)

Background
In 2019, cardiovascular disease (CVD) was the leading global cause of disease burden, affecting approximately 523 million individuals. Prior research has established that individuals of lower socioeconomic status (SES) are more likely to endure illness and early death than persons of higher SES. Still, there is no clear answer to which SES metric is the best measure of cardiovascular health, in low- and middle-income countries.

Design
Members of the Health and Aging in Africa: A Longitudinal Study of an INDEPTH Community in South Africa (HAALSI) baseline population were used for this analysis. Participants who were eligible for this study provided information on hypertension and diabetes in addition to the household asset and consumption measures (N=4,953/5,059; 97.9%).

Methods
Log-Binomial regression models with robust standard errors and logarithmic-link functions (tailored for binary outcomes) were used to generate prevalence ratios (PRs) and the 95% confidence intervals (CIs). Adjusted models accounted for age, sex, country of birth, self-rated health during childhood, education, literacy, and employment status. The first quintile (Q1; the poorest and lowest asset/consumption class) were used as the reference categories.

Results
There is a positive relationship between household assets and consumption explaining hypertension and diabetes among our study population.

Conclusion
This study is an example of how researchers can quantify an SES metric unique to the population of interest, such as household asset and consumption quintiles, when uncovering the relationships between exposures and outcomes. More research is needed to evaluate the causal implications surrounding this finding using the same HAALSI population.

Keywords: South Africa · Aging · Rural · Socioeconomics · Inequalities · Hypertension · Diabetes · Prevalence

INTRODUCTION

In 2019, cardiovascular disease (CVD) was the leading global cause of disability, and disease burden, affecting approximately 523 million individuals.(1,2) By 2023, the World Health Organization (WHO) and Institute for Health Metrics and Evaluation (IHME) predicted a cumulative 23.6 million total deaths worldwide due to CVD.(8,9) Although a large proportion of global CVD mortality is estimated to occur in low and middle-income countries, the role of socioeconomic status (SES) is still undefined when assessing CVD in these settings.(10,11)

Previous research has established that individuals of lower SES are more likely to endure illness and early death than persons of higher SES.(3) More specifically, low SES has consistently been correlated with recognized CVD risk factors, such as high blood pressure, among adult populations.(9,10,12–14) Studies have measured SES as a social or occupational class, educational attainment, income, house size, household consumption, and ownership of certain household assets, as reflected in a 'wealth' index.(3)

Researchers in 2020 randomly selected a sample of 2,300 men in eastern Finland and their self-reported income, education, occupational prestige, material standard of living, along with housing conditions to create a summary index of SES that ranged from 0 to 25; with high values assigned to lower SES participants.(4) This type of indexing guided the impact of cardiorespiratory fitness (CRF) on the survival of men with low SES. They uncovered that SES and CRF are independently associated with all-cause and CVD mortality among that population.(4) In alternative to generating index scores, researchers opted to use income quintiles to survey the prevalence and odds of having a stroke or angina in low and middle-income countries.(13) Using this form of reporting income, study investigators determined that higher household income is associated with lowered odds of angina in countries like China, India, and the Russian Federation.(13) However, the opposite effect was observed in South Africa.(13)

Although household income is commonly used as an indicator for population health status, especially in child health, there are still challenges in gathering reliable income data through surveys in low- and middle-income countries.(7) Participant's lack of stable employment, in consideration of their age, has prompted researchers to explore the use of household consumption and asset measures as indicators in countries whose populations have inconsistent income reports.(5,7) Investigators compared household wealth with consumption metrics when determining different health standings among older South African adults in rural communities.(7) The baseline data used in this assessment was from a 2015 longitudinal study called the Health and Aging in Africa: A Longitudinal Study of an INDEPTH Community in South Africa (HAALSI).(15) Findings from the study revealed that income data, in supplement to consumption data, can produce a clearer appearance of inequalities within socioeconomics and health of older persons.(7)

To build on the application of household consumption, scholars used the same HAALSI baseline data to quantify the link between household wealth, household consumption, and chronic disease risk behaviors among older South African adults in rural communities.(5,15) They have uncovered an increased household consumption was associated with low moderate-to-vigorous intensity physical activity. Higher consumption was also associated with higher BMI levels, smoking, and alcohol use.(5)

Still, there is no clear answer to which SES metric is the best measure of health, specifically cardiovascular health, in low- and middle-income countries.(4) Nonetheless, it is still important to explore the intersection between SES and CVD to provide greater clarity on a

person's lifestyle, and influence for creating and evaluating health interventions or policies.(1,2,7) Moreover, since older adults experience growing health-related issues, there is a need to increase the literature on cardiometabolic health in older adult populations.(16)

Therefore, the objective of this study will be to investigate the association between socioeconomic positions, as measured by indices of household assets and consumption, concerning the outcomes of hypertension and diabetes among adults aged ≥ 40 in rural Agincourt sub-district, South Africa. The purpose of this research is to formally quantify SES within this particular cohort population as a significant factor for those living with cardiometabolic diseases.(3) Given the findings from previous literature, we hypothesize that increased household consumption and assets will be protective against cardiometabolic diseases such as hypertension and diabetes among members of this baseline population.

METHODS

Study Population

The data was collected from baseline assessments in 2014-2015 in the "Health and Aging in Africa: A Longitudinal Study of An INDEPTH Community in South Africa" (HAALSI). The HAALSI study is a population-based study of 5,059 randomly sampled men and women aged ≥ 40 years from the Agincourt Health and Social Demographic Surveillance System (AHDSS). The AHDSS sampling frame included an estimated 116,000 people living in 31 villages within the Agincourt sub-district, Mpumalanga province, South Africa (response rate = 86%).(5,15) This aging population had to endure decades of the HIV/AIDS epidemic, rising chronic disease risk behaviors, and poor access to health management.(5,10,15,16) There is a history of systemic racism, lack of developed infrastructure, low access to education and employment, alongside poor economic opportunities in the Agincourt region.(5) Participants who were eligible for the present analysis were those who successfully provided information on hypertension and diabetes in addition to the exposure and covariate measures (N=4,953/5,059; 97.9%).

Data Collection

Interviews were conducted in person with local and skilled fieldworkers in the participants' homes in the Shangan language and using computer-assisted personal interviews (CAPI). The interview questions were translated into the specific language of the participants and then back-translated from English for quality assurance. The interview topics included but were not limited to sociodemographic elements, health status, social networks, risky health behaviors, and living conditions.(5,15)

Measures

Exposure Variables: Quintiles of Household Assets and Consumption

An asset-based household wealth index was constructed from interview data on housing conditions, durable and non-durable assets, land ownership, and livestock ownership.(5) Participants were assigned to quintiles of the household asset index variable for descriptive statistics and statistical models. Household per capita consumption was calculated by combining data on regular and non-regular expenses of different food products, goods, and services in addition to food made at home.(5) Participants were then assigned to quintiles of the consumption variable for descriptive statistics and statistical models. This consumption variable is considered a more reliable measure for living standards in developing countries because it

provides a holistic picture of the household than what is produced or purchased by household members.(5,7) Both household measures were scaled based on the number of persons living in the home.(7,15)

Outcome Variables: Cardiometabolic Diseases

In this study, CVD is characterized as hypertension and diabetes. Hypertension was measured as participants providing a mean systolic blood pressure ≥ 140 mmHg, a mean diastolic blood pressure of ≥ 90 mmHg, self-reporting diagnosis of hypertension validated by a medical professional, or self-reported use of medication for hypertension prescribed by a medical professional.(17) Diabetes among study participants was also measured as participants having a fasting plasma glucose ≥ 7.0 mmol/L, a random plasma glucose ≥ 11.1 mmol/L, self-reporting diagnosis of diabetes validated by a medical professional, or self-reported use of medication for diabetes prescribed by a medical professional.(17)

Covariates

Demographic variables such as sex (male or female), age (continuous), and country of birth (South Africa, Mozambique, or Other) were included as covariates in this analysis. Additional variables included highest educational attainment (no formal education; some primary education [1–7 years]; some secondary education [8–11 years]; secondary education or higher [12 or more years]), current employment status (not employed, working part-time or full time, or homemaker), literacy (self-reported ability to read or write; yes/no) and self-reported health in childhood (very bad, bad, moderate, or good, very good).(15) Continuous age was used instead of categorical age values to encompass all reported age values participants listed. Furthermore, the inclusion of "country of birth" as a variable was incorporated to describe any unmeasured differences based on whether people were born in South Africa or persons who migrated to the Agincourt area during the 1977-1992 civil war in Mozambique, a neighboring country.(18)

Statistical Analysis

Descriptive statistics were used to understand the distribution of exposures, covariates, and the prevalence of outcomes in the study population. We used Log-Binomial regression models with robust standard errors and logarithmic-link functions (tailored for binary outcomes) to generate prevalence ratios (PRs) and the 95% confidence intervals (CIs) for the relationships between each household asset and consumption quintiles with diabetes and hypertension.(5,6) Adjusted models accounted for age, sex, country of birth, self-rated health during childhood, education, literacy, and employment status. The first quintile (Q1; the poorest and lowest asset/consumption class) for both household categories were used as the reference categories. The analyses were conducted using SAS version 9.3 (SAS Institute Inc., Cary, NC, USA).

RESULTS

Missing Data

No data were missing on age, sex, household assets, or household consumption variables. There were 106/5059 (2%) missing data on hypertension and 407/5059 (8%) missing data on diabetes. The variables for education (17/5059), literacy (3/5059), childhood health (4/5059), employment status (14/5059) had less than 1% of missing data. The analytical sample for the

generalized estimating equation (GEE) regression involved all participants with complete data on the exposure, outcome, and covariates (N=4,953/5,059; 97.9%).

Results

Table 1 provides the baseline characteristics of all the participants in this sample. As an example, 58% of women were born in South Africa, 42% could not read nor write, and 46% had no formal education. There were 26% of participants working part-time, full-time, or as a homemaker, with a vast majority of participants self-reporting having good or very good childhood health (88%). Lastly, the average age of participants in the sample was 62 years.

Table 1: *Characteristics of the sample, "Health and Aging in Africa: A Longitude Study of an INDEPTH Community in South Africa"* (HAALSI), Agincourt, South Africa, 2014-15.

Characteristics of the sample	**N (%) 5059 (100%)**
Sex	
Male	2345 (46)
Female	2714 (53.65)
Age, $\bar{x}$ (sd)	62 (13)
Country of Birth	
South Africa	3528 (69.8)
Mozambique or Other	1526 (30.2)
Literacy	
Can read and/or write	2948 (58.3)
Cannot read or write	2108 (41.7)
Education Level	
No Formal Education	2306 (45.7)
Some Primary (1-7 Years)	1716 (34)
Some Secondary (8-11 Years)	574 (11.4)
Secondary or More (12+ Years)	446 (8.9)
Employment Status	
Not Employed	3719 (73.7)
Working Part-Time or Full-Time	805 (15.9)
Homemaker	521 (10.3)
Self-Rated Childhood Health	
Good or Very Good	4434 (87.7)
Moderate, Bad, or Very Bad	621 (12.3)
Household Assets Quintiles	
Quintile 1 (Lowest):	1046 (20.7)
Quintile 2:	1001 (19.8)
Quintile 3:	991 (19.6)
Quintile 4:	1007 (19.9)
Quintile 5 (Highest):	1014 (20.0)
Household Consumption Quintiles	
Quintile 1 (Lowest):	1047 (20.7)
Quintile 2:	1024 (20.2)
Quintile 3:	1021 (20.2)
Quintile 4:	975 (19.3)

Quintile 5 (Highest):	992 (19.6)
Diabetes	559 (12.2)
Hypertension	3145 (63.5)

As household assets and consumption quintiles increased, the prevalence of diabetes in the population increased as well. The participants in asset quintiles first (lowest) and second had a 12.7% and 16.1% prevalence of diabetes. The participants in asset quintiles fourth and fifth (highest) had 23.1% and 27.5% prevalence of diabetes, respectively (Figure 1). There was a 14.8%-point difference in diabetes prevalence between participants in the first and fifth quintiles of household assets (Figure 1). Similarly, the prevalence of diabetes among people in the first quintile of household consumption was 16.3%, and the prevalence of diabetes among people in the fifth quintile was 26.5%, a 10.2%-point difference (Figure 1).

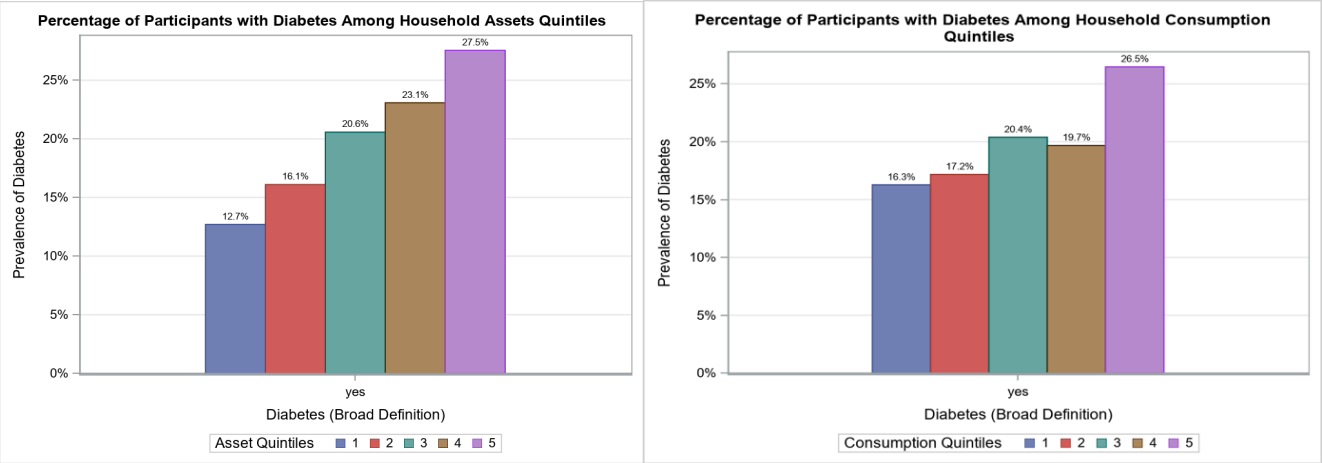


Figure 1: Prevalence of Diabetes in Study Population. *The histograms depict the prevalence of diabetes in the study population based on their position in the household asset quintile (left) and in the household consumption quintile (right). "Health and Aging in Africa: A Longitude Study of an INDEPTH Community in South Africa" (HAALSI), Agincourt, South Africa, 2014-15, N= 4,653.*

Conversely, the prevalence of hypertension was similar across household assets and consumption quintiles (Figure 2). On average, approximately 20% of participants within each consumption and asset quintile had hypertension (Figure 2). People in the first quintile of household assets had a prevalence of 18.7%, while people in the fifth quartile had a 21.4% prevalence of hypertension, a 2.7%-point difference (Figure 2). On the other hand, the prevalence of hypertension among people in the first quintile of household consumption is 19.6%, and the prevalence of hypertension among people in the fifth quintile is 26.5%. As a result, there is a 6.9%-point difference in hypertension between those groups of participants (Figure 2).

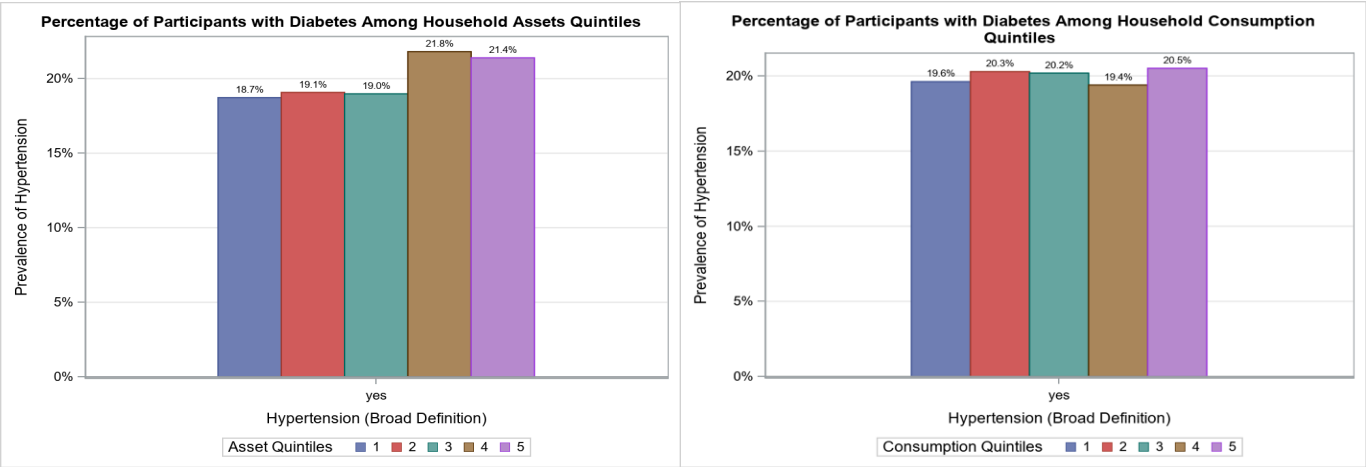


Figure 2: Prevalence of Hypertension in Study Population. *The histograms depict the prevalence of hypertension in the study population based on their position in the household asset quintile (left) and in the household consumption quintile (right). "Health and Aging in Africa: A Longitude Study of an INDEPTH Community in South Africa" (HAALSI), Agincourt, South Africa, 2014-15, N= 4,953.*

Unadjusted prevalence ratios (PR) reported diabetes was 1.32 times higher (95% CI = 0.97, 1.79) among those in the second household assets quintile compared to those in the first household assets quintile (Table 2). The PR increased from the first quintile to the fifth quintile, where the prevalence of diabetes was 2.21 times higher (95% CI = 1.69, 2.90) compared to those in the lowest household assets quintile (Table 2). The unadjusted PR associating diabetes and consumption indicated that diabetes prevalence was 1.08 times higher (95% CI = 0.78, 1.50) among those in the second household consumption quintile compared to those in the first household consumption quintile (Table 2). The PR increased from the lowest quintile to the highest quintile, where the prevalence of diabetes is 1.73 times higher (95% CI = 1.34, 2.23) compared to those in the first household consumption quintile (Table 2).

The unadjusted PR revealed that the prevalence of hypertension is 1.06 times higher (95% CI = 0.99, 1.13) among those in the second household asset quintile compared to those in the first household assets quintile (Table 2). The PR increased from the lowest quintile to the highest quintile, where the prevalence of hypertension is 1.17 times higher (95% CI = 1.09, 1.26) compared to those in the first household assets quintile (Table 2). Unadjusted PR between consumption quintiles explaining hypertension signifies the prevalence of the CVD being 1.06 times higher (95% CI = 0.99, 1.14) among those in the second household consumption quintile compared to individuals in the first household consumption quintile (Table 2). The PR increased from the lowest quintile to the highest quintile, where the prevalence of hypertension is 1.1 times higher (95% CI = 1.02, 1.19) compared to those in the first household consumption quintile (Table 2).

Table 2: Unadjusted prevalence ratios (PRs) for hypertension and diabetes, according to household asset and consumption quintiles. *"Health and Aging in Africa: A Longitude Study of an INDEPTH Community in South Africa"* (HAALSI), Agincourt, South Africa, 2014-15.

Variables	**Diabetes**		**Hypertension**	
Household Asset Quintiles	PR	95% CI*	PR	95% CI*
Quintile 1 (Lowest)	1.00(ref)		1.00(ref)	
Quintile 2	1.32	(0.97, 1.79)	1.06	
Quintile 3	1.70	(1.37, 2.11)	1.07	(0.98, 1.16)
Quintile 4	1.89	(1.50, 2.38)	1.21	(1.12, 1.30)
Quintile 5 (Highest)	2.21	(1.69, 2.90)	1.17	(1.09, 1.26)
Household Consumption Quintiles				
Quintile 1 (Lowest)	1.00(ref)		1.00(ref)	
Quintile 2	1.08	(0.78, 1.50)	1.06	(0.99, 1.14)
Quintile 3	1.29	(0.93, 1.79)	1.05	(0.98, 1.13)
Quintile 4	1.32	(1.04, 1.68)	1.07	(0.99, 1.16)
Quintile 5 (Highest)	1.73	(1.34, 2.23)	1.10	(1.02, 1.19)

**95% Confidence Intervals were calculated using Walds Chi-squared statistics.*

Adjusted PR revealed the prevalence of diabetes is 1.26 times higher (95% CI = 0.93, 1.72) among those in the second household assets quintile compared to those in the first household assets quintile (Table 3). The PR increased from the lowest quintile to the highest quintile, where the prevalence of diabetes is 1.99 times higher (95% CI = 1.47, 2.68) compared to those in the first household assets quintile (Table 3). Adjusted PR, connecting diabetes and consumption, states the prevalence of diabetes is 1.01 times higher (95% CI = 0.72, 1.41) among those in the second household consumption quintile compared to those in the first household consumption quintile (Table 3). The PR increased from the lowest quintile to the highest quintile, where the prevalence of diabetes is 1.47 times higher (95% CI = 1.09, 2.00) compared to those in the first household consumption quintile (Table 3).

Adjusted PR informed the prevalence of hypertension is 1.02 times higher (95% CI = 0.97, 1.07) among those in the second household asset quintile compared to those in the first household assets quintile (Table 3). The PR increased from the lowest quintile to the highest quintile, where the prevalence of hypertension is 1.09 times higher (95% CI = 1.04, 1.15) compared to those in the first household assets quintile (Table 3). Adjusted PR linking hypertension and consumption highlights the prevalence of hypertension is 1.03 times higher (95% CI = 0.99, 1.08) among those in the second household consumption quintile compared to those in the first household consumption quintile (Table 3). The PR increased from the lowest quintile to the highest quintile, where the prevalence of hypertension is 1.06 times higher (95% CI = 1.01, 1.12) compared to those in the first household consumption quintile (Table 3).

Table 3: Adjusted prevalence ratios of cardiometabolic diseases, such as hypertension and diabetes, according to household asset and consumption quintiles. *"Health and Aging in Africa: A Longitudinal Study of an INDEPTH Community in South Africa"* (HAALSI), Agincourt, South Africa, 2014-15.

Variables	Diabetes		Hypertension	
Household Asset Quintiles	PR	95% CI*	PR	95% CI*
Quintile 1 (Lowest)	1.00(ref)		1.00(ref)	
Quintile 2	1.26	(0.93, 1.72)	1.02	(0.97, 1.07)
Quintile 3	1.57	(1.26, 1.96)	1.03	(0.98, 1.08)
Quintile 4	1.68	(1.33, 2.12)	1.10	(1.05, 1.15)
Quintile 5 (Highest)	1.99	(1.47, 2.68)	1.09	(1.04, 1.15)
Household Consumption Quintiles				
Quintile 1 (Lowest)	1.00(ref)		1.00(ref)	
Quintile 2	1.01	(0.72, 1.41)	1.03	(0.99, 1.08)
Quintile 3	1.17	(0.84, 1.63)	1.02	(0.97, 1.07)
Quintile 4	1.18	(0.90, 1.54)	1.02	(0.98, 1.08)
Quintile 5 (Highest)	1.47	(1.09, 2.00)	1.06	(1.01, 1.12)

**95% Confidence Intervals were calculated using Walds Chi-squared statistics.*

DISCUSSION

The purpose of this research is to determine if there was a relationship between the prevalence of diabetes and hypertension, given household assets and consumption gradients in this particular cohort.(3) However, we have found a positive relationship between household gradients explaining CVD among our study population. The conclusions drawn from the statistical analysis refuted our hypothesis that increased household assets and consumption habits are protective against diabetes and hypertension.

These findings contradict studies that suggest an inverse relationship between SES and risk of cardiometabolic disease in different populations across the globe. A study conducted in 2019 had an objective to examine the association between change in household income and subsequent risk of CVD.(19) Using an ongoing prospective cohort study of men and women who live in four United States (U.S.) communities, researchers quantified that income drop over six years was associated with a higher risk of CVD incidence over 17 years. Meanwhile, income rise over six years was associated with a lower risk of CVD incidence over 17 years.(19)

In Columbia, a study that utilized a nationally representative survey of adults older than 60 years of age also concluded that individuals with lower SES have a significantly higher risk of developing high blood pressure (HBP) than individuals of higher SES. Researchers revealed an SES gradient in HBP matched to additional risk factors for CVD based on study findings.(20)

A third study, conducted in rural and peri-urban India, sought to assess the prevalence of cardiometabolic risk factors by socioeconomic position (SEP) among adults.(21) From their analysis, researchers concluded that risk factors did not show a consistent association with SEP but had a higher prevalence in activities such as smoking and alcohol intake among lower SEP groups.(21)

Though these projects were based on data collected from the U.S., Columbia, and India research sites, all three studies represent a population-wide gradient of highest, medium, and lowest income countries that assess some form of SES concerning developing CVD or cardiometabolic risk factors. Nonetheless, our study reflects the opposite effect in the previously stated literature. Individuals who fall within higher household quintiles are also the individuals

with a higher prevalence of hypertension and diabetes than study participants residing only in the first quintile.

When considering the outcome, more participants had hypertension then diabetes (hypertension = 63.5%; diabetes = 12.02%). We suspect there to be participants who had both hypertension and diabetes. Likewise, there are reasons to suspect increased screening of the outcome of hypertension among individuals taking medication for CVD or HIV.(16,17) There are reasons to suspect a mediation effect between individuals with sexually transmitted infections (STI) and the prevalence of diabetes and hypertension, given the increased screening procedures for individuals who frequently visit the clinic or hospital for health services.(16,17) Furthermore, there is a chance that individuals in low-income countries with more assets and higher consumption habits will increase their likelihood of diabetes or hypertension due to the abundance of resources they hold versus their lower quintile counterparts. This phenomenon is still being evaluated from studies that personally investigate the relationship between income-based metrics, assets or consumption, and the onset of CVD in low-income countries.

There are several limitations in this study. Due to analyzing baseline measurements, the study design reflects a cross-sectional examination of a prospective cohort study. Thus does not establish causality between the exposure causing the outcome.(5,16) Moreover, the nature of this study design does not measure the risk for diabetes and hypertension in this population. Therefore, we were unable to draw those conclusions given the parameters of the study analysis. Lastly, the study results are not exchangeable to other groups or populations outside of the baseline HAALSI study participants. This critique prevents our efforts from being fully reproducible in populations not derived from the HAALSI cohort study. Further investigations are needed to confirm these findings in longitudinal form of this cohort study. Likewise, there is a need to evaluate the effectiveness of this model for researching SES gradients concerning the prevalence of diabetes and hypertension.

Since an increase in household asset and consumption quintiles was directly proportional to the increase in the prevalence ratios, these findings refute the study hypotheses and the literature presented in prior years. More research is needed to evaluate the causal implications surrounding this finding using the same HAALSI population. This study is an example of how researchers can quantify an SES metric unique to the population of interest, such as household asset and consumption quintiles, when uncovering the relationships between exposures and outcomes.(5,7) Future research can aid in filling the knowledge gap of how income is measured for low-income communities around the world regarding chronic or infectious disease outcomes.

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